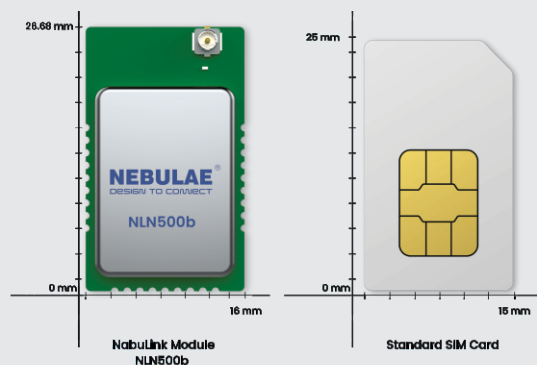


NLN500b

The NebuLink Module NLN 500b is smallest STM32WL based LoRa Module. It is build on the STM32WL SoC which has Embedded LoRaWAN Transceiver for wide range of Smart Applications. It is useful when long range is required as it operates at 865/868 MHz band and uses LoRa (Long-Range) and Sigfox Protocol for communication. It offers long range, low power and secure data transmission for M2M and IoT applications. This Module designed to be extremely low-power and is based on the high-performance Arm® Cortex® - M4 32-bit RISC core operating at a frequency of up to 48 Mhz.



Specifications

Performance

TX Power		+21dBm
RX Sensitivity	865Mhz, BW=125Khz	DR0/SF12, - 136 dBm
	865Mhz, BW=125Khz	DR5/SF7, - 123 dBm
RF Range (Line of Sight)		More than 10km
RF Data Rate		DR0-DR5

Processor Specifications

Processor	Core: 32-bit Arm®Cortex®-M4 CPU, frequency up to 48 MHz, Up to 43 I/Os
Memory	Up to 256KB flash, 64KB SRAM
Frequency Band	EU 863-870, US 902-928, IN865
Mounting Type	1.27 mm Pitch SMD
Antenna Option	u.FL, Spring Antenna, Feedline
Size (LxWxH in mm)	26.68x16.0x3.6

IO Specifications

Serial Interface	2 x U(S)ART 1 x LPUART
Analog Input	9 x 12-bit ADC 2.5 Msps
Digital IO	Up to 21 pins

Electrical Specifications

Supply Voltage(VDD)	1.8V~3.6V, 3.3V
Current Consumption, 865-867 Mhz	
(PA= +21dBm), TX Mode	146mA
(PA= +21dBm), RX Mode	8 mA
(PA= +13dBm), TX Mode	106mA
(PA= +13dBm), RX Mode	8mA
Operating Temperature	-30° C to 85° C

Networking

Protocol	LoRaWAN® and Sigfox™
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Applications



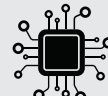
Smart
Metering



Smart
Agriculture



Industrial
Automation



M2M &
IoT

Key Product Features

- Frequency Band: Supports EU 863-870, IN865
- RX Sensitivity: -123 dBm (125 kHz BW and spreading factor 7) -136 dBm (125kHz BW and spreading factor 12) for LoRa
- Transmitter Output Power: Up to +22 dBm
- Supported Protocols: LoRaWAN® and Sigfox™
- Power Supply: 1.8V to 3.6 VDC
- Operating Temperature Range: -20°C to +75°C
- Core: 32-bit Arm®Cortex®-M4 CPU
- Hardware Encryption: AES 256-bit
- Memory: Up to 256-Kbyte Flash memory, Up to 64-Kbyte RAM
- Update: OTA (over-the-air) firmware update capable
- Programming/Debug: Serial-wire debug (SWD), JTAG
- I/Os: up to 21 I/Os
- I/O Tolerance Voltage: 5VDC
- Directly soldered on any device board
- RF shield on the module

Contact us System Level Solutions

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